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Pathology-interpretable radiomic model for predicting clinical outcome in patients with osteosarcoma: a retrospective, multicenter study

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Abstract

Background Osteosarcoma is the most prevalent primary malignant bone tumor. Radiomic models show promise in globally evaluating the prognosis of osteosarcoma; however, they lack biological interpretability. We aimed to develop a radiomic model using MRI to predict disease-free survival (DFS) in osteosarcoma patients and provide the underlying pathobiology of the model.

Methods This retrospective study included 270 patients (training set, $n=166$; external test set 1, $n=56$; external test set 2, $n=48$) with surgically treated and histology-proven osteosarcoma from 14 tertiary centers. A total of 1130 radiomic features were extracted from baseline MRI data. After dimensionality reduction, the radiomic model was developed on the training set and tested on the external test sets. The radiomics interpretability study used the hematoxylin and eosin (H&E) and immunohistochemistry (IHC) stained whole slide images (WSIs) of patients from the testing sets. Ten types of nuclear morphological features were extracted from each nucleus in H&E WSIs and aggregated into 150 patient-level features. Furthermore, five immune- and hypoxia-related IHC biomarkers, including CD3, CD8, CD68, FOXP3, and CAIX, were quantified from IHC WSIs. The correlation between the radiomic features and histopathologic markers was assessed using Spearman or Pearson correlation analysis, with multiple hypothesis testing controlled by the false discovery rate.

Results The radiomic model comprising 12 features yielded a time-dependent AUC of 0.916 (95% CI: 0.893–0.939), 0.802 (95% CI: 0.763–0.840), and 0.895 (95% CI: 0.869–0.920) in the training set, external test set 1, and external test set 2, respectively. Notably, nine out of 12 radiomic features exhibited significant correlations with 17 cellular features,

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resulting in 32 pairs. Specifically, there were four (12.5%) pairs with absolute coefficient *r* (|*r*|) between 0.3 and 0.4, 22 (68.8%) pairs between 0.4 and 0.5, and six (18.8%) pairs exceeding 0.5. Four radiomic features were correlated with CD3 ($r = 0.50–0.75$), two features with CD8 ($r = 0.46$ and 0.60), and three features with CD8/FOXP3 ($r = 0.69–0.81$).

Conclusions The MRI-based radiomic model shows potential in predicting DFS in osteosarcoma patients. Most radiomic features show only moderate associations with H&E-derived nuclear morphological features; they exhibit higher correlations with immune-related biomarkers.

Keywords Osteosarcoma, Magnetic resonance imaging, Radiomics, Hematoxylin and eosin, Immunohistochemistry, Disease-free survival

Background

Osteosarcoma, a primary malignant bone tumor that typically arises in the metaphyseal regions near growth plates, is the most common and aggressive bone malignancy in children and adolescents [1]. Despite its relatively low incidence, with age-standardized rates of 2.9 per 1,000,000 for males and 2.2 per 1,000,000 for females, its highly invasive nature and tendency for metastasis pose significant therapeutic challenges [2]. The standard treatment paradigm, involving neoadjuvant chemotherapy (NAC) followed by surgery and adjuvant therapies, has markedly improved survival outcomes [3]. Yet, a notable proportion of patients with localized osteosarcoma experience recurrence risks, with 5-year survival rates of 23–29% [4]. Thus, identifying novel biomarkers for disease progression is critical to advance osteosarcoma treatment. Developing a biomarker-based model for early prediction of disease progression can inform personalized treatment decisions, thereby improving individual patient survival rates.

Magnetic resonance imaging (MRI) has become an indispensable tool for staging, therapeutic response evaluation, and posttreatment surveillance of primary bone tumors. MRI also facilitates personalized treatment prediction and monitoring, serving as an indicator of treatment response or prognosis [5]. While advanced MRI techniques have improved preoperative diagnostic accuracy, the reliance on radiologists' expertise and the limitations of MRI sequences in capturing tumor heterogeneity highlight the need for precise and objective imaging biomarkers. In recent years, radiomic analysis, involving the high-throughput extraction of quantitative image features, has shown potential in predicting the response to NAC in osteosarcoma [6–8]. Several studies also have shown that the radiomic models outperformed clinical models in predicting the osteosarcoma progression [9–11]. Nevertheless, these studies have been limited to single-center or dual-center settings, lacking independent validation of model generalizability.

While radiomic models depend on many computational features, the biological meaning of these data-driven features often remains elusive, which can hinder model interpretability [12]. A connection between quantitative imaging features and biological processes through interdisciplinary technologies will bridge the gap between imaging data and tumor biology, making the models more transparent and actionable [13]. In addition, biological validation of radiomic features ensures that the model captures meaningful tumor characteristics rather than imaging artifacts or noise. Recently, the biological rationale of radiomics has been explored through genomic analysis, histopathological methods, or other experimental assays [12]. Among these, histopathological data is more readily accessible in clinical practice. Hematoxylin and eosin (H&E) and immunohistochemistry (IHC) stained whole slide images (WSIs) provide valuable insights into the tumor microenvironment (TME), including tissue architecture, cellular morphology, as well as hypoxia and immune factors [14, 15]. Previous studies have elucidated the correlation between histology-depicted TME and clinical outcomes [16–18]. Integrating radiomics with digital pathology can enhance the interpretability of the former by linking imaging features with pathobiology. This approach not only strengthens the reliability of radiomic models but also offers clinicians more understandable predictive insights.

In this current study, we aimed to develop and validate a radiomic model for predicting individual disease-free survival (DFS) in patients with osteosarcoma using baseline MRI data. The performance of the radiomic model was benchmarked against the conventional clinical model, with its robustness evaluated across independent test sets. In addition, we also interpreted the biological underpinnings of radiomic models using pathological features extracted from H&E- and IHC-stained WSIs.

Methods

Ethics approval

This study was approved by the ethics committee of the First Affiliated Hospital of Jinan University (approval number: 20230227). Informed consent was waived because of the retrospective nature of the study and the analysis used anonymous clinical and pathological data.

Study design

The overall study design is outlined in Fig. 1. We developed and validated an MRI-based radiomic model to predict DFS in osteosarcoma patients using multicenter datasets. The generalization ability of the radiomic model was validated on two independent external test sets. This work was carried out following the Methodological Radiomics Score (METRICS), a quality scoring tool for radiomics research endorsed by EuSoMII [19]. In addition, we investigated the biological significance of radiomic features using H&E- and IHC-stained WSIs from patients in the external test sets. Ten types of cell-level morphological features were automatically extracted from each nucleus in H&E WSIs and aggregated into a total of 150 patient-level features. Furthermore, hypoxia-related and immune-related TME biomarkers were obtained through IHC analysis. Spearman's correlation analysis was conducted to evaluate the relationship between radiomic features and histopathological features.

Patient cohort and data acquisition

This retrospective study screened 896 osteosarcoma patients from 14 tertiary centers between January 2004 and January 2024. Inclusion criteria were as follows: (1) patients with histopathologically confirmed osteosarcoma based on the surgical specimen; (2) patients who underwent surgery with or without NAC; (3) patients who underwent contrast-enhanced MRI scans 1 or 2 weeks prior to treatment; (4) patients who had not received prior antitumor therapy before MRI examinations; and (5) patients without concurrent malignancies. Exclusion criteria were as follows: (1) patients lacking contrast-enhanced MRI images or with insufficient image quality (e.g., artifacts, low signal-to-noise ratio, and low resolution); (2) patients with less than 6 months of follow-up without disease progression. Finally, 270 patients were included. Figure 2 displays the patient recruitment pathway.

The distribution of sample sizes across different centers was as follows: center 1 ($n=120$), center 2 ($n=34$), center 3 ($n=12$), center 4 ($n=56$), center 5 ($n=10$), center 6

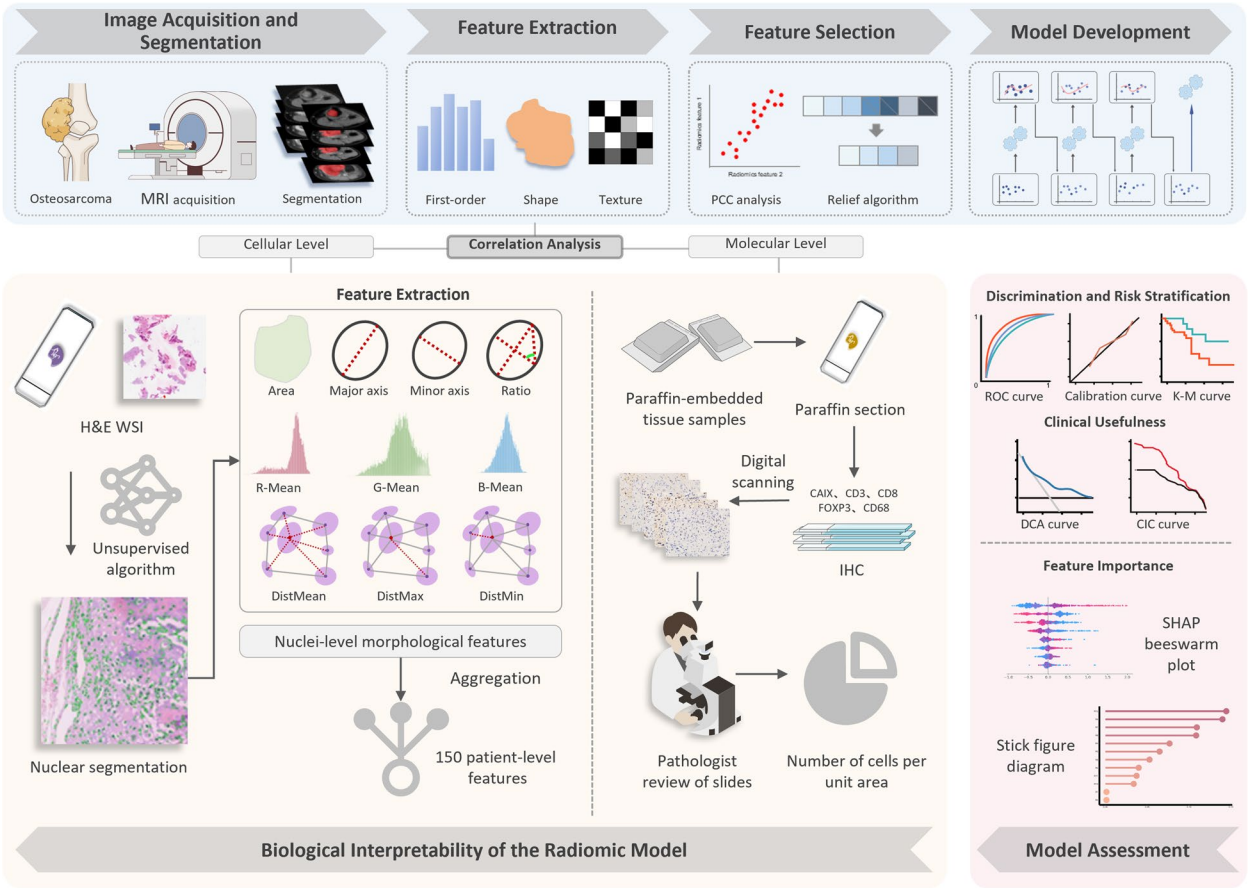


Fig. 1 Overall study design. The main steps include MR image acquisition and segmentation, feature extraction, feature selection, model development, model assessment, as well as biological interpretability of the radiomic model

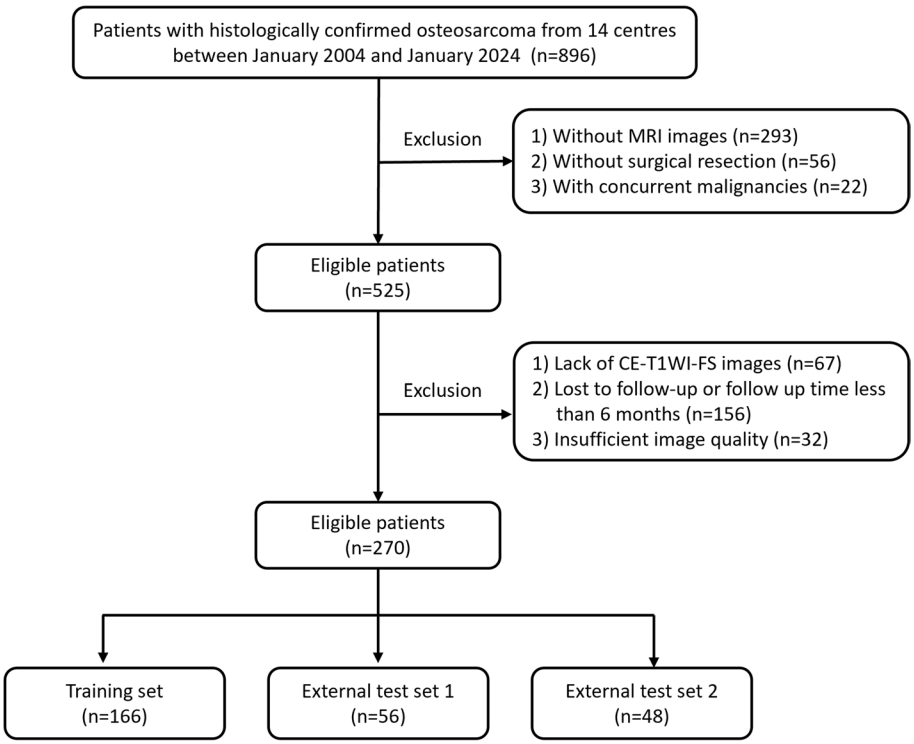


Fig. 2 Patient inclusion and exclusion flowchart

($n=9$), center 7 ($n=8$), center 8 ($n=6$), center 9 ($n=5$), center 10 ($n=5$), center 11 ($n=2$), center 12 ($n=1$), center 13 ($n=1$), and center 14 ($n=1$). The division of different datasets was based on the geographical distribution of centers. The training set for developing the predictive model included 166 patients from centers 1 to 3 (101 males and 65 females; mean age, 21.1 years ± 13.2). The external test set 1 comprised 56 patients from center 4 (39 males and 17 females; mean age, 21.4 years ± 11.5), while the external test set 2 comprised 48 patients from centers 5 to 14 (27 males and 21 females; mean age, 21.4 years ± 12.0).

Baseline clinical, radiological, and pathological data were collected, including age, gender, tumor location, tumor stage, maximum tumor diameter, tumor volume, presence of pathological fracture, and pathological subtypes. All images were acquired from the Picture Archiving and Communication System. Detailed MRI acquisition parameters are presented in the Additional file 1: Table S1. Tumor volume was calculated as follows: $V=\pi/6\times L\times W\times H$, where L, W, and H represent the length, width, and height of the tumor measured on MRI, respectively [20]. The tumor was staged according to the American Joint Committee on Cancer (AJCC) Staging System for Bone Tumors (Eighth Edition) [21]. The presence of pathological fracture at diagnosis was determined by consensus among three experienced radiologists (with over 10 years of experience). The study endpoint was DFS, which was defined as the time from the date of definitive surgery to disease progression or death. For patients with metastatic lesions, they underwent complete resection of both the primary tumors and the metastatic lesions (primarily lung metastases). The starting point for DFS calculation was the date of the last metastatic lesion resection to ensure uniform post-surgical follow-up. Data collection and evaluation were completed in January 2024.

MRI image normalization, tumor segmentation, and radiomic feature extraction

Contrast-enhanced fat-suppressed T1-weighted imaging (CE-FS-T1WI) provides sufficient spatial and contrast resolution to capture tumor heterogeneity while maintaining model simplicity. CE-FS-T1WI is a widely used sequence in clinical practice for evaluating musculoskeletal tumors, delivering excellent contrast between the tumor and surrounding tissues, which is essential for accurate tumor segmentation and feature extraction [22]. Integrating multi-parametric MRI data across 14 centers with heterogeneous protocols would have introduced technical variability, potentially undermining feature stability and model generalizability

[23]. Additionally, several centers had missing MRI sequences other than CE-FS-T1WI. To maximize the inclusion of cases, we selected the CE-FS-T1WI, which was available across all centers. The MRI images were processed using the N4 bias correction algorithm in 3D Slicer software to mitigate MRI field distortions caused by radiofrequency field inhomogeneity and inherent MR equipment effects. A grayscale normalization algorithm was applied to standardize MRI intensity values to [0, 255], reducing grayscale variations among MRI images from different patients, acquisition times, and parameter settings. Finally, the voxels of the ROI were resampled using a cubic spline interpolation algorithm (1*1*1 mm^3), ensuring standardized analysis.

Tumors in the training set were independently segmented by two radiologists (each with 5 years of experience in musculoskeletal imaging) using ITK-SNAP software [24]. All regions of interest (ROI) were manually delineated slice by slice to encompass each gross tumor volume on CE-FS-T1WI. Due to potential inter-observer variability in tumor segmentation, segmentations were evaluated by a senior radiologist specializing in musculoskeletal imaging with 15 years of experience. Final segmentation in the test sets was produced by a single reader to better reflect clinical practice. To assess segmentation variability, we randomly selected 50 cases from the training set and evaluated the segmentation results from the two radiologists. We used the Dice Similarity Coefficient (Dice) to quantify the overlap between their segmentations and the Intraclass Correlation Coefficient (ICC) to evaluate the reliability of the radiomic features extracted from the segmented regions.

Using the open-source Pyradiomics v3.0 package, radiomic features were extracted from original images and preprocessed images using the Laplacian of Gaussian (LoG) filter and wavelet transformation. The majority of these features adhere to the definitions outlined by the Imaging Biomarker Standardization Initiative (IBSI) [25]. They encompass four categories: (1) shape-based features ($n=14$); (2) intensity histogram-based features ($n=72$); (3) matrix texture features ($n=300$), including the gray-level co-occurrence matrix (GLCM), gray-level run length matrix (GLRLM), gray-level size zone matrix (GLSZM), gray-level dependence matrix (GLDM), and neighborhood gray-tone difference matrix (NGTDM); and (4) wavelet features ($n=744$). A total of 1130 radiomic features were extracted from CE-FS-T1WI (see Additional file 1: Table S2 for specific feature names).

Radiomic model development

Prior to feature selection, radiomic feature values were normalized to [0, 1] using the MinMax algorithm. Subsequently, Pearson correlation analysis was applied to initially identify a minimally redundant feature set (with a correlation coefficient threshold set at 0.9). Further feature selection was performed using the Relief algorithm. Various machine learning classifiers, including Random Forest (RF), Support Vector Machine (SVM), Logistic Regression (LR), Adaptive Boosting (AdaBoost), Multilayer Perceptron (MLP), and Naive Bayes (NB), were applied. Fivefold cross-validation was utilized for feature selection and algorithm optimization to construct the top-performing radiomic model. The entire procedure was conducted on the training set, while the external test sets were used exclusively for model validation. Standardization parameters obtained from the training set were applied to preprocess the independent test set data to evaluate the model's generalization performance.

The potential association of radiomic signature (i.e., Radscore) and DFS was assessed in the training set and then validated in the external test sets 1 and 2 using Kaplan–Meier survival analysis. The patients were divided into high- and low-risk subgroups according to the predictive Radscores; the threshold of which was identified by using X-tile software. The difference in the survival curves of the high- and low-risk subgroups was evaluated by using a weighted log-rank test.

Clinical models were established by employing univariate and multivariate Cox proportional hazards regression analysis to screen clinical characteristics. Likewise, utilizing the multivariate Cox regression method, the combined model was constructed by integrating significant clinical features and Radscores. The predictive performance of the radiomic model was compared with that of the clinical model, tumor volume, and combined model.

Radiology-pathology correlation analysis: Interpretability of the radiomic model

H&E WSI-derived morphological nuclear features

We collected 41 patients from the external test set 1, each with available H&E-stained slides of the osteosarcoma. The slides were digitalized into SVS format using the Digital Pathology Slide Scanner KF-PRO-400 (Ningbo, Zhejiang, China) at a 40× magnification. All the WSIs were manually checked by a senior pathologist for artifacts, and the images free of all types of artifacts were chosen. We pre-processed each WSI with a pixel threshold to filter out a white background, where pixels with a mean value of the RGB channels greater than 210 were considered as background. We then divided all the non-background areas into non-overlapping 2048*2048 patches and leveraged an unsupervised segmentation algorithm [26] to segment cell nuclei on

the patches. For each segmented nucleus, we extracted 10 types of morphological nuclear features, including nuclear area, lengths of the major and minor axes of the cell nucleus, the ratio of major axis length to minor axis length (major, minor, and ratio), mean pixel values of the nucleus in RGB three channels, and mean, maximum, and minimum distances (distant, distMax, and distMin) to its neighboring nuclei in the Delaunay triangulation graph [27] (Additional file 1: Table S3). Finally, for each type of morphological nuclear feature, a 10-bin histogram and five statistical measurements (i.e., mean, standard deviation, skewness, kurtosis, and entropy) were used to aggregate the cell-level features and generate a 150-dimensional imaging feature for each patient (Additional file 1: Table S4). All feature extraction was implemented using MATLAB R2021a.

IHC-derived hypoxia-immune-related biomarkers

Forty patients from the external test sets with available formalin-fixed paraffin-embedded (FFPE) tissue samples were included. FFPE samples were cut into 3-mm thick sections, which were then processed for IHC staining (Appendix file 2: Protocol of IHC-staining). The detected immune-related and hypoxia-related IHC biomarkers included CD3 (pan T cells), CD8 (cytotoxic T cells), CD68 (macrophages), FOXP3 (Treg cells), and Carbonic Anhydrase IX (CAIX) (Additional file 3: Fig. S1). IHC evaluation was performed independently by two pathologists (with 5 and 10 years of experience, respectively) who were blinded to the outcome data. In cases of disagreement between the primary pathologists, a third pathologist was consulted to reach a consensus. In detail, the tissue sections were initially screened at low power (100×) using an upright microscope (BX53; Olympus, Japan), and the five most representative fields were selected. The density of immune and hypoxic cells was then assessed at 200×magnification within tumor regions. The cells stained for nuclear (FOXP3) and membrane (CD3, CD8, CD68, and CAIX) biomarkers were quantified and expressed as the number of cells per unit area (cells/mm^2).

Correlation analysis

The correlation between the radiomic and histopathologic features was evaluated using Spearman or Pearson correlation analysis, with multiple hypothesis testing controlled by the false discovery rate. The histopathologic features comprised 150 patient-level cellular morphological features derived from H&E WSIs and five hypoxia-immune-related biomarkers derived from IHC. Correlation coefficients below 0.3 were considered weak correlations, 0.3 to 0.7 as moderate correlations, and > 0.7 as strong correlations [28].

Statistical analysis

Statistical analyses were conducted using R version 4.4.0 and Python 3.7.6. The patient characteristics among the three datasets were compared using the Kruskal–Wallis test for continuous variables and the chi-square test or Fisher's exact test for categorical variables. The predictive performance of the radiomic model was evaluated using time-dependent receiver operating characteristic (ROC) curves, calibration curves, decision curve analysis (DCA), and clinical impact curve (CIC). Time-dependent ROC analysis was conducted using the "timeROC" package. The DeLong test was employed to compare the area under the curve (AUC) between models. The calibration curve was generated using the "rms" package. The Hosmer–Lemeshow test was carried out using the "ResourceSelection" package to examine the goodness of fit between model-predicted and observed probability. The DCA and CIC were plotted using the "rmda" package in R. The DCA illustrates the net benefit of using the predictive models compared to other strategies (such as always treating or never treating) at various probability thresholds [29]. The CIC shows the potential impact of adopting the predictive model on clinical decision outcomes across various probability thresholds [30]. The Shapley Additive Explanations (SHAP) algorithm was used to visually quantify the contribution of each feature within the radiomic model using "shap" package [31]. Statistical significance was set at a P value < 0.05.

Role of funding source

The funders played no role in the study design, data collection, data analysis, data interpretation, and writing of the report.

Results

Patient characteristics

This retrospective study included 270 patients, with their baseline characteristics summarized in Table 1. The cohort was divided into three non-overlapping groups: the training set comprised 166 patients (median age: 17 years; 60.8% male), external test set 1 included 56 patients (median age: 18 years; 69.6% male), and external test set 2 consisted of 48 patients (median age: 17 years; 56.3% male). There were differences in pathologic fracture, pathological subtype, tumor size, tumor volume, and the use of neoadjuvant chemotherapy among the three datasets (all P < 0.05).

Evaluation of manual-segmentation variability

The Dice similarity coefficients yielded a median value of 0.859 (interquartile range [IQR]: 0.758–0.898), underscoring the high consistency and reliability of our

Table 1 Clinical characteristics of patients in the training and testing sets

Characteristics	Training set (*n* = 166)	External test set 1 (*n* = 56)	External test set 2 (*n* = 48)	*P*-value
Age (median [IQR], years)	17.0 [13.0, 24.0]	18.0 [14.0, 23.0]	17.0 [13.0, 23.0]	0.732
Gender				0.341
Female	65 (39.2)	17 (30.4)	21 (43.8)	
Male	101 (60.8)	39 (69.6)	27 (56.3)	
Tumor location				0.198
Long bone	122 (73.49)	47 (83.93)	40 (83.3)	
Flat bone	44 (26.51)	9 (16.07)	8 (16.7)	
Pathologic fracture				0.022
No	151 (91.0)	50 (89.3)	37 (77.1)	
Yes	15 (9.0)	6 (10.7)	11 (22.9)	
Pathological subtype[a]				0.008
Conventional	129 (77.7)	53 (94.6)	41 (85.4)	
Others	37 (22.3)	3 (5.4)	7 (14.6)	
Tumor size (median [IQR], cm)	8.3 [6.8, 11.0]	7.8 [6.6, 9.6]	6.8 [5.7, 8.4]	< 0.001
Tumor volume (median [IQR], cm^3)	203.8 [95.3, 377.1]	256.5 [156.0, 421.3]	210.9 [95.4, 367.2]	0.013
T stage				< 0.001
T1	76 (45.8)	31 (55.4)	35 (72.9)	
T2	87 (52.4)	25 (44.6)	12 (25.0)	
T3	3 (1.8)	0	1 (2.1)	
N stage				0.463
N0	165 (99.4)	56 (100)	48 (100)	
N1	1 (0.6)	0	0	
M stage				0.189
M0	155 (93.4)	49 (87.5)	46 (95.8)	
M1[b]	11 (6.6)	7 (12.5)	2 (4.2)	
Neoadjuvant chemotherapy[c]				< 0.001
Yes	156 (94.0)	8 (14.3)	28 (58.3)	
No	6 (3.6)	47 (83.9)	13 (27.1)	
Unknown	4 (2.4)	1 (1.8)	7 (14.6)	

Unless otherwise specified, data are numbers of patients, with percentages in parentheses

IQR, interquartile range

[a] Conventional osteosarcoma is the most common type, comprising various subtypes such as osteoblastic, chondroblastic, fibroblastic, and fibrohistiocytic osteosarcoma

[b] All distant metastases were lung metastases

[c] Some patients received a minimum of 4–6 cycles (i.e., 2–3 months) of NAC before undergoing surgical resection, adhering to the National Comprehensive Cancer Network guidelines

segmentation. Also, the ICCs demonstrated a median value of 0.924 (IQR: 0.788–0.977), indicating robust feature extraction from the segmented regions. Specifically, the results showed that 908 features achieved ICCs of 0.75 or higher, 835 features reached ICCs of 0.80 or higher, and 634 features attained ICCs of 0.90 or higher. Collectively, these findings suggest the strong robustness and consistency of the majority of the radiomic features derived from the segmented regions. Additionally, we have included the visualization results in Additional file 3: Fig. S2.

Predictive performance of the radiomic model

After performing dimensionality reduction, we identified 12 radiomic features significantly associated with clinical outcomes, comprising four first-order features and eight textural features (Additional file 1: Table S5). Comparison of six machine learning algorithms using cross-validation revealed that the AdaBoost-based radiomic model achieved the highest performance (Fig. 3a–c), with AUC values of 0.916 (95% CI: 0.893–0.939), 0.802 (95% CI: 0.763–0.840), and 0.895 (95% CI: 0.869–0.920) in the training set, external test set 1, and external test set 2, respectively.

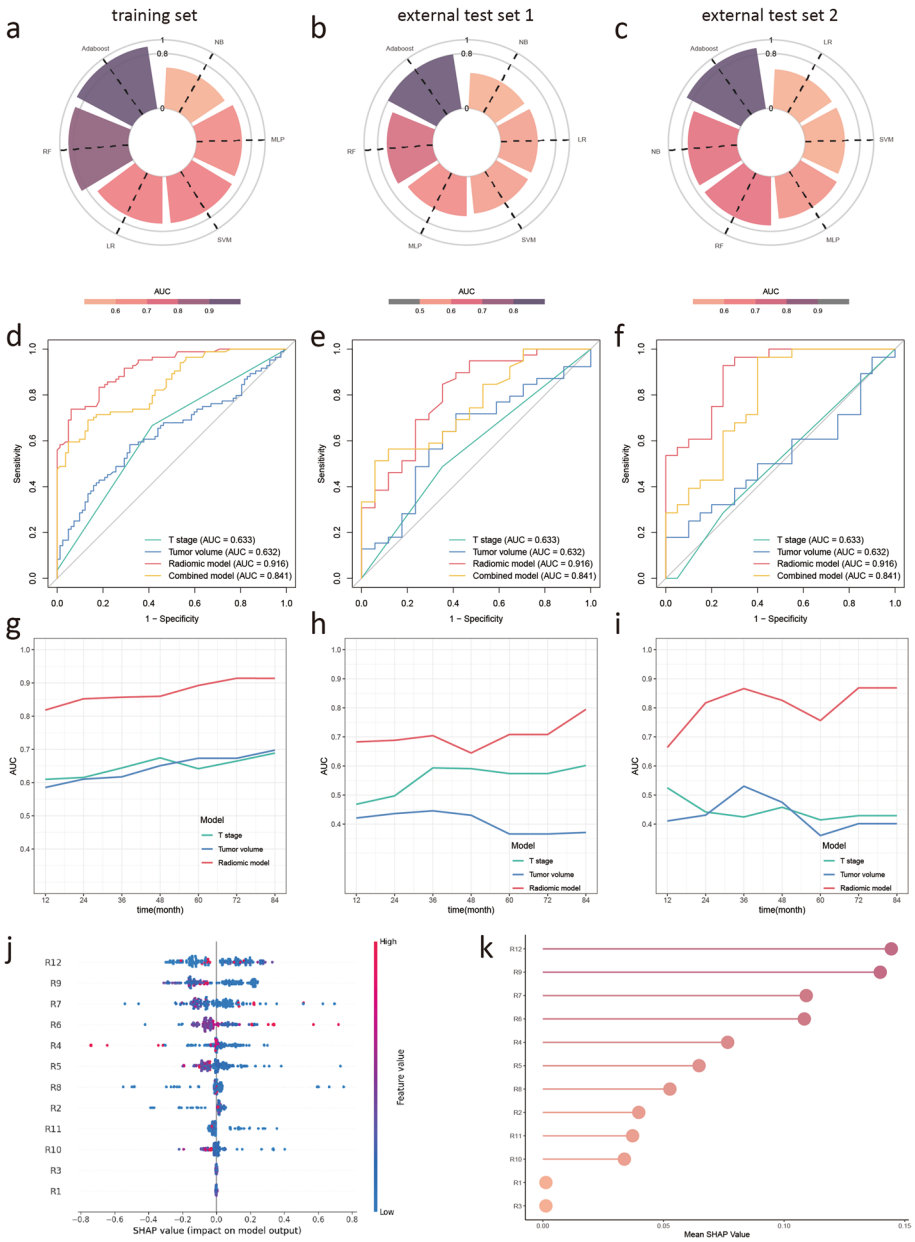


Fig. 3 The radar plot displays the AUC values for six radiomics-based machine learning algorithms in both training and testing sets (**a**–**c**). The ROC curves for the T stage, tumor volume, radiomic model, and combined model in training and testing cohorts are shown (**d**–**f**). The radiomic model showed higher AUC values at most time points compared to tumor T stage and tumor volume. Time-dependent AUCs were measured annually from 1 to 7 years, reflecting prediction performance at various time points. The top-performing model (radiomic model) is compared with tumor T stage and tumor volume in both training and testing sets (**g**–**i**). The SHAP beeswarm plot illustrates the importance of radiomic features (**j**). In this plot, the *x*-axis represents each feature's impact on model prediction according to its SHAP value, with the absolute value indicating the magnitude of the impact and positive/negative values showing increases/decreases in the predicted probability. A bar graph ranks variables by their importance to the model, based on the mean of the absolute values of all Shapley values for 12 radiomic features (**k**). Abbreviations: AUC, area under the curve; ROC, receiver operating characteristic; SHAP, Shapley Additive exPlanations. R1: original_firstorder_Entropy; R2: original_firstorder_InterquartileRange; R3: original_glcm_Contrast; R4: original_glrlm_ShortRunLowGrayLevelEmphasis; R5: original_glszm_ZonePercentage; R6: wavelet_HHL_glcm_ClusterShade; R7: wavelet_HLH_firstorder_Variance; R8: wavelet_HLL_glcm_ClusterProminence; R9: wavelet_LHL_firstorder_Variance; R10: wavelet_LHL_glcm_DifferenceVariance; R11: wavelet_LLL_glszm_LargeAreaEmphasis; R12: wavelet_LLL_glszm_LargeAreaHighGrayLevelEmphasis

Tumor volume is well recognized as a critical prognostic factor in solid tumors. We evaluated its predictive accuracy for DFS in osteosarcoma patients, with time-dependent AUC values of 0.632 (95% CI: 0.582–0.681), 0.623 (95% CI: 0.575–0.671), and 0.527 (95% CI: 0.480–0.574) in the training set, external test set 1, and external test set 2, respectively (Table 2 and Fig. 3d–f). Following Cox analyses of clinical variables, only tumor T stage emerged as an independent risk factor (Additional file 1: Table S6), yielding AUC values of 0.633 (95% CI: 0.591–0.676), 0.567 (95% CI: 0.525–0.609), and 0.511 (95% CI: 0.474–0.548) in the same sets (Table 2 and Fig. 3d–f). Hence, a combined model integrating tumor T stage and Radscore achieved AUC values of 0.841 (95% CI: 0.808–0.875), 0.756 (95% CI: 0.717–0.794), and 0.789 (95% CI: 0.751–0.828) in these respective sets (Table 2 and Fig. 3d–f). In comparison, Delong tests indicated that the radiomic model significantly outperformed tumor T stage, tumor volume, and the combined model (all $P<0.05$). Thus, the T stage did not contribute additional predictive value beyond the radiomic model. The dynamic change of model performance over time further validated these findings (Fig. 3g–i). Figure 3j–k illustrates the SHAP values, reflecting the contributions of different radiomic features to the model's prediction.

The radiomic model could classify patients into high- and low-risk subgroups with distinct survival probabilities based on a threshold of 0.49 (Fig. 4a–c). The calibration curves (Fig. 4d–f) show good agreement between radiomic model-predicted DFS probabilities and actual DFS probabilities (Hosmer–Lemeshow P values = 0.596, 0.581, 0.317, respectively). This indicates that our model's predictions were reliable and could reflect the true DFS probabilities. The decision curves show that the radiomic model achieves a superior net benefit compared to the T stage and tumor volume (Fig. 4g–i). The CICs show that the radiomic model provided a superior overall net benefit within the wide ranges (65–100%) of threshold probabilities and impacted patient outcomes in both the training and test sets (Fig. 4j–l).

The reporting and methodological quality of this present study was evaluated by the METRICS tool [32], which includes 30 items across 9 categories. The total METRICS score of our study was 93.7%, suggesting excellent quality (Additional file 3: Fig. S2).

Radiopathomic correlation analysis: Interpretability of the radiomic model

To elucidate the underlying pathobiology of radiomic features, we conducted a Spearman correlation analysis involving 150 nuclear morphological features and 12 top-predictive radiomic features (Fig. 5a). Notably, nine out of the 12 radiomic features showed significant correlations with 17 (11.3%) nuclear morphological features, yielding 32 correlation pairs. In detail, there were four (12.5%) pairs with $|r|$ between 0.3 and 0.4, 22 (68.8%) pairs with $|r|$ between 0.4 and 0.5, and six (18.8%) pairs with $|r|$ exceeding 0.5 (up to 0.56) (Fig. 5b). Figure 5c–d depicts the count and proportion of nuclear feature types within each radiomic feature. The chord plot shows significant radiology-pathology correlations (all $P<0.05$). Specifically, the correlogram (Fig. 5e) shows 32 correlation pairs with $|r|>0.3$ between nine radiomic features and 17 pathological features.

To deepen our understanding of how the radiomic features relate to TME, we performed Pearson correlation analyses with IHC-derived hypoxia-immune-related biomarkers (Fig. 5f). The results demonstrate varying degrees of correlations between the radiomic features and these biomarkers. Overall, there were nine pairs

Table 2 Predictive performance of the models

Dataset	Model	AUC (95% CI)	*P*-value	Sensitivity (%)	Specificity (%)	Accuracy (%)
Training set	Tumor T stage	0.633 (0.591–0.676)	< 0.001	66.7	58.5	62.7
	Tumor volume	0.632 (0.582–0.681)	< 0.001	58.3	67.1	62.7
	Radiomic model	0.916 (0.893–0.939)	Ref	91.7	70.7	81.3
	Combined model	0.841 (0.808–0.875)	< 0.001	75.0	59.8	67.5
External test set 1	Tumor T stage	0.567 (0.525–0.609)	< 0.001	48.7	64.7	53.6
	Tumor volume	0.623 (0.575–0.671)	< 0.001	53.8	70.6	58.9
	Radiomic model	0.802 (0.763–0.840)	Ref	79.5	64.7	75.0
	Combined model	0.756 (0.717–0.794)	0.031	71.8	52.9	66.1
External test set 2	Tumor T stage	0.511 (0.474–0.548)	< 0.001	28.6	75.0	47.9
	Tumor volume	0.527 (0.480–0.574)	< 0.001	50.0	55.0	52.1
	Radiomic model	0.895 (0.869–0.920)	Ref	96.4	60.0	81.3
	Combined model	0.789 (0.751–0.828)	< 0.001	67.9	65.0	66.7

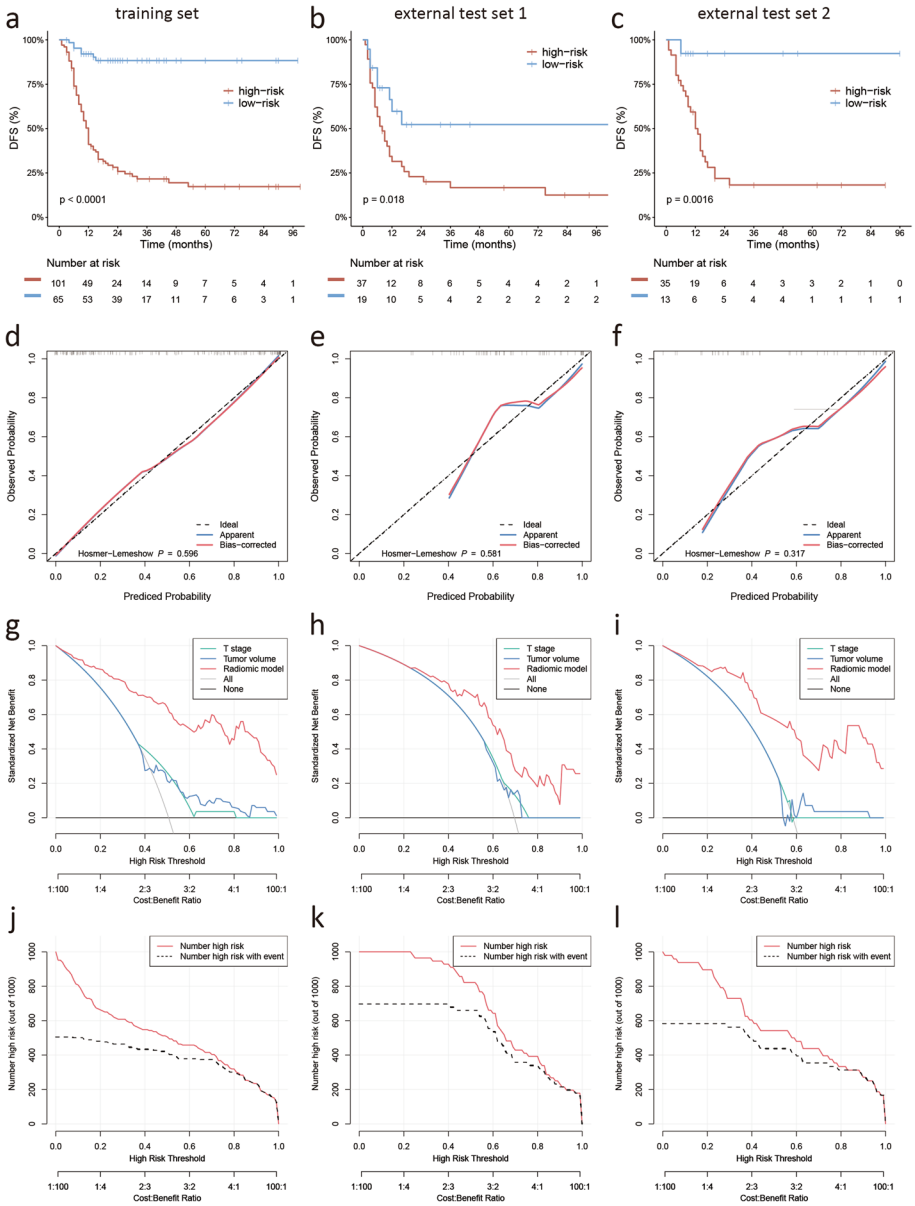


Fig. 4 The Kaplan–Meier curves illustrate DFS probabilities for patients in the low- and high-risk groups across the training and testing sets (**a**–**c**). The calibration plots display the apparent calibration curve and the bias-corrected calibration curve after bootstrapping for both the training and testing sets (**d**–**f**). Decision curve analysis for T stage, tumor volume, and the radiomic model is shown for the training and testing sets (**g**–**i**). The *y*-axis measures the net benefit, while the *x*-axis represents the predictive probability threshold. The CIC indicated that the that the radiomic model achieved high true positive rates and cost-effectiveness across a wide range of risk thresholds in both the training and testing sets (**j**–**l**). Abbreviations: DFS, disease-free survival; DCA, decision curve analysis; CIC, clinical impact curve. R1: original_firstorder_Entropy; R2: original_firstorder_InterquartileRange; R3: original_glcm_Contrast; R4: original_glrlm_ShortRunLowGrayLevelEmphasis; R5: original_glszm_ZonePercentage; R6: wavelet_HHL_glcm_ClusterShade; R7: wavelet_HLH_firstorder_Variance; R8: wavelet_HLL_glcm_ClusterProminence; R9: wavelet_LHL_firstorder_Variance; R10: wavelet_LHL_glcm_DifferenceVariance; R11: wavelet_LLL_glszm_LargeAreaEmphasis; R12: wavelet_LLL_glszm_LargeAreaHighGrayLevelEmphasis

showing positive correlations ($r = 0.46$–0.81) (Fig. 5g). Specifically, four features with CD3 ($r = 0.50$–0.76), two features with CD8 ($r = 0.46$–0.60), and three features with CD8/FOXP3 ($r = 0.69$–0.81). Notably, there were six (66.7%) correlation pairs with r values ranging from 0.3 to 0.7 (moderate correlations), and three pairs (33.3%) with *r* values exceeding 0.7 (strong correlations).

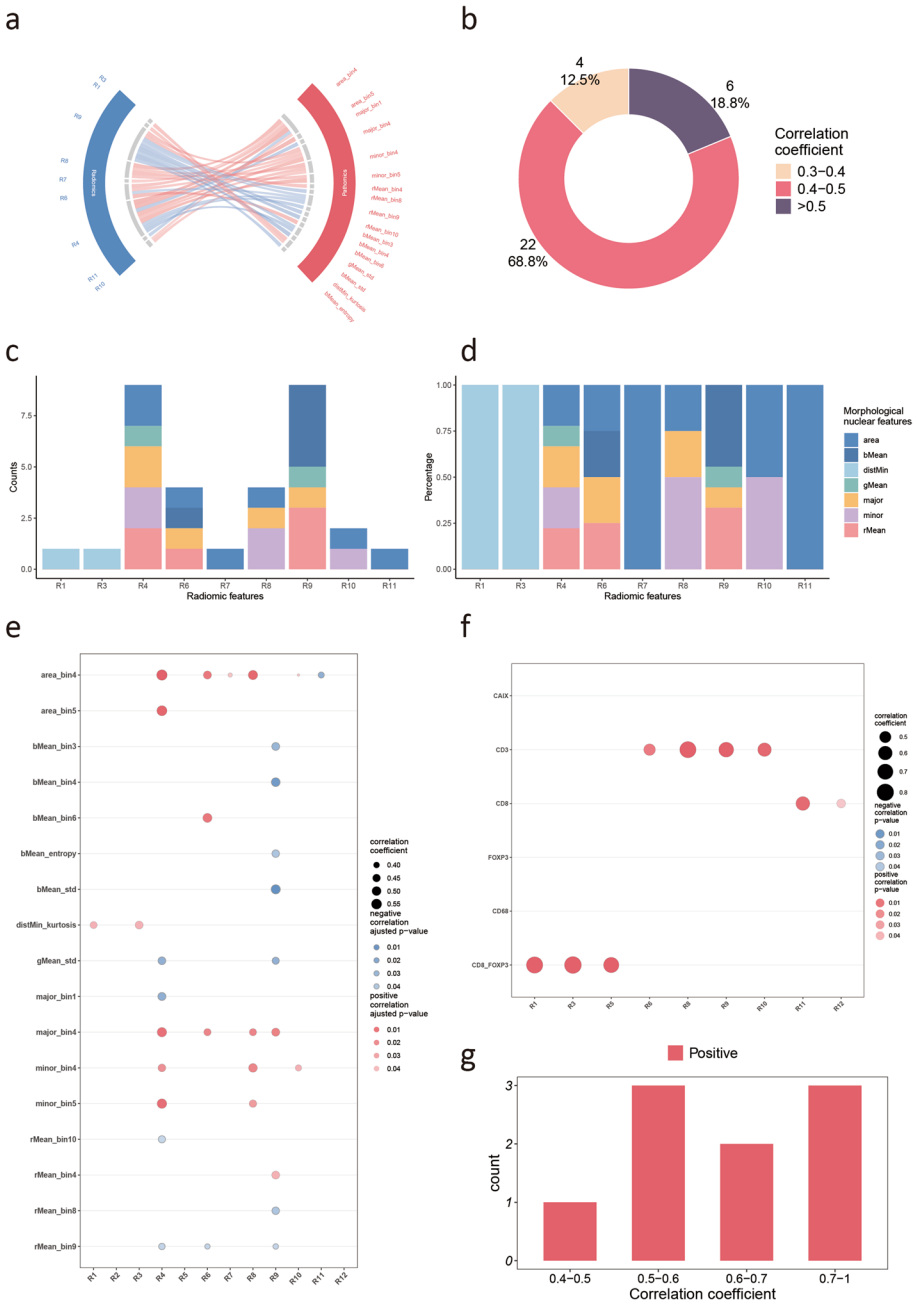


Fig. 5 Chord diagram of significant correlations ($P < 0.05$) between radiomic features and nuclear features (**a**). Line color indicates correlation levels: red for positive correlation and blue for negative correlation. The number and percentage of correlation coefficients across different strength intervals among 32 pairs are shown (**b**), with the size of each colored section representing the percentage of the correlation coefficient range. The stacked bar charts illustrate the distribution of ten types of nuclear morphological features within each radiomic feature, presented as count (**c**) and percentage (**d**) views. Bubble plots illustrate the correlation between radiomic features and nuclear features (**e**), as well as between radiomic features and the IHC-derived TME biomarkers (**f**). Bubble size represents correlation levels, and color denotes *P* value levels: red for positive correlation and blue for negative correlation. The bi-directional bar graph displays the distribution of correlation coefficients with intervals of 0.1 (**g**). Red: positive correlation; Blue: negative correlation. R1: original_firstorder_Entropy; R2: original_firstorder_InterquartileRange; R3: original_glcm_Contrast; R4: original_glrlm_ShortRunLowGrayLevelEmphasis; R5: original_glszm_ZonePercentage; R6: wavelet_HHL_glcm_ClusterShade; R7: wavelet_HLH_firstorder_Variance; R8: wavelet_HLL_glcm_ClusterProminence; R9: wavelet_LHL_firstorder_Variance; R10: wavelet_LHL_glcm_DifferenceVariance; R11: wavelet_LLL_glszm_LargeAreaEmphasis; R12: wavelet_LLL_glszm_LargeAreaHighGrayLevelEmphasis

Discussion

In this study, we developed and validated an MRI-based radiomic model that could predict DFS in patients with osteosarcoma, which outperformed empirical clinical markers such as tumor T stage and tumor volume. Furthermore, we constructed a cross-scale correlation framework between macroscopic radiomic features and microscopic digital pathology-based TME features. The results demonstrated that most of the radiomic features correlated with cellular morphological features extracted from H&E-stained WSIs, with 32 (11.7%) pairs exhibiting moderate correlations. However, the correlations were stronger between the radiomic features and immune-related biomarkers, including CD3, CD8, and CD8/FOXP3 ratio.

Radiological imaging allows for noninvasive, comprehensive evaluation of the entire tumor and can be repeated as needed throughout the treatment course. Radiomics refers to the process of extracting quantitative data from routine medical images, which can capture the inherent heterogeneity within tumors [33]. This approach involves advanced computational analysis techniques that transform standard medical images into mineable data [34]. By extracting a large number of quantitative features from images, radiomics allows for a more global characterization of tumor heterogeneity than traditional visual or qualitative assessments alone [35]. Understanding and quantifying tumor heterogeneity through radiomics can provide valuable insights into tumor biology, prognosis, and treatment response. Previous radiomic studies have employed mainly treatment responses to NAC and prognosis prediction in patients with osteosarcoma [6–11]. However, most studies lacked external validation and had small sample sizes due to the rarity of osteosarcoma (around 100), which could lead to an overestimation of model predictive accuracy or overfitting. Improvements in study design and validation are necessary to show the generalizability of findings and to facilitate clinical applications.

To our knowledge, our study represents the largest sample size in previous research on MRI radiomics predicting treatment response and prognosis in osteosarcoma. Our study demonstrates that the imaging-based radiomic model containing four first-order and eight texture features significantly outperformed clinical factors such as T stage and tumor volume, showing superior prognostic accuracy. Given its complementary value, this model has the potential to enhance the precision of the current staging system and improve risk stratification for osteosarcoma. One strength of our study is the validation of the prognostic model across diverse populations. Given that the choice of classifiers directly impacts the predictive performance of radiomic models, we compared six machine learning classifiers and ultimately selected AdaBoost. AdaBoost is an ensemble learning method that combines multiple weak classifiers to create a strong classifier [36]. It focuses on the most challenging data points by iteratively adjusting the weights of misclassified instances. This, along with its simplicity and minimal hyperparameter tuning requirements, makes AdaBoost highly effective for complex classification tasks. In our study, these likely contributed to AdaBoost's superior performance compared to other models. However, we should acknowledge that AdaBoost can be sensitive to noisy data and may overfit if the number of iterations is too large. To mitigate these potential biases, we employed fivefold cross-validation for algorithm optimization to construct the top-performing radiomic model. We noted variability in performance across the test sets, which may be attributed to differences in clinical variables (e.g., sample size and T-stage) and scanning protocols (e.g., field strengths and acquisition parameters). The radiomic model's clinical utility lies in its ability to stratify patients into distinct prognostic subgroups, thereby personalizing therapeutic strategies. For example, high-risk patients could be prioritized for adjuvant immunotherapy or targeted therapies, while low-risk patients might benefit from reduced chemotherapy exposure. However, integrating the radiomic model into the clinical workflow needs multidisciplinary collaboration.

We utilized SHAP values to elucidate the contributions of radiomic features to the model's predictions. The most important features were all derived from wavelet-transformed analysis. For instance, the top-1 feature wavelet_LLL_glszm_LargeAreaHighGrayLevelEmphasis reflects the presence of large, homogeneous regions with high signal intensity, which may indicate areas of necrosis, calcification, or dense tumor tissue. This feature may be associated with tumor aggressiveness or therapy response. The top-2 and top-3 features wavelet_LHL_firstorder_Variance and wavelet_HLH_firstorder_Variance indicate the variance in the wavelet-transformed image (LHL and HLH bands), capturing the variability in pixel intensity. These features also reflect tumor heterogeneity but focus on a different spatial frequency, potentially highlighting fine-textured patterns or subtle variations in tumor structure. The top-4 feature wavelet_HHL_glcm_ClusterShade derives from the GLCM and measures the asymmetry of the intensity distribution in the HHL wavelet band. Cluster Shade quantifies the skewness of the GLCM, reflecting the presence of irregular or asymmetric texture patterns. Higher values may indicate more complex or disorganized tumor structures. This feature could be

linked to tumor invasiveness or metastatic potential, as disorganized tissue patterns often correlate with aggressive tumor biology. However, the clinical relevance of the radiomic features remains theoretically inferred and warrants further in-depth investigation.

A significant obstacle to the clinical translation of current data-driven radiomic models is their lack of interpretability, often stemming from a disconnect from the underlying pathobiology [13]. The absence of biological interpretability in the model leads to our inability to intuitively understand the mechanisms behind the predictions. Although previous studies [37–41] have utilized transcriptomic data to decipher the biological functions of radiomic features, compared to other omics data such as transcriptomics and proteomics, pathological data is biologically closer to imaging data, which may suggest potentially stronger correlations between them. Moreover, pathological images contain rich information regarding the biological behavior of tumors, with robust evidence showing that nuclear heterogeneity is closely linked to tumor prognosis [42]. H&E images provide information about tissue architecture and cellular morphology, which is crucial for understanding the underlying biology of tumors. It accurately detects, grades, and evaluates clinically relevant findings for tumors, and can provide new insights into the relationship between tumor profiles and prognosis [43]. IHC is a powerful tool for identifying and quantifying specific proteins or markers within tissues. Through extensive literature review and careful consideration, we focused on hypoxia-immune-related biomarkers, including CD3, CD8, CD68, FOXP3, and CAIX, which play significant roles in the TME of osteosarcoma. These markers provide insights into the hypoxia-immune landscape of the tumor, which is critical for understanding prognosis and potential therapeutic targets [44, 45]. In our study, we extracted 10 types of morphological features of tumor cell nuclei to quantitatively evaluate nuclear heterogeneity and then aggregated them into 150 patient-level features. We studied the correlation between the significant radiomic features and nuclear morphology, and the results showed that most of the radiomic features exhibited varying degrees of significant correlation with nuclear morphological features, which reflects that nuclear heterogeneity to some extent determines the heterogeneity of imaging. Nine radiomic features exhibited correlations > 0.30 with 17 (11.3%) nuclear features. The moderate correlation between radiomic and pathological features may result from multiple factors, such as scale discrepancy, sampling bias, localization of immune infiltrates, and difficult image-pathology registration.

Accumulating evidence suggests that TME has a nonnegligible role in tumorigenesis, proliferation, metastasis, and drug resistance acquisition in osteosarcoma, thereby affecting the prognosis of patients with osteosarcoma [46–48]. In addition, low immune cell infiltration levels in TME not only represent lower anti-tumor immunity but also help tumor cells to evade immune attacks [49, 50]. Stromal cells, such as cancer-associated fibroblasts, exert a direct immunosuppressive effect [50, 51]. This suggests that the TME status may be a potential prognostic biomarker as well as a predictive marker for the response to treatments such as immunotherapy in osteosarcoma. The prognostic significance of the TME has been well established across various solid tumors. The TME of osteosarcoma exhibits highly immunosuppressive properties and is often described as an immune "cold" tumor [52]. Shi et al. found varying levels of exhaustion in $CD8^+$ T cells in the osteosarcoma microenvironment [53]. The intratumoral accumulation of $FOXP3^+$ Tregs has been identified as a major immune evasion mechanism in various tumors, including osteosarcoma [54]. Fritzsching et al. [55] showed that osteosarcoma patients with higher intratumoral $CD8^+/FOXP3^+$ ratios had better outcomes compared to those with lower ratios. We found significant correlations between radiomic features and CD3, CD8, as well as the CD8/FOXP3 ratio. The high correlations suggest that radiomics could non-invasively identify candidates for emerging immunotherapies (e.g., anti-PD1/CTLA-4), predict responses to macrophage-targeting agents (e.g., mifamurtide) and neoadjuvant regimens, and facilitate prognostic risk stratification. Apart from the immune microenvironment, hypoxia is also an inherent feature of osteosarcoma, which is related to tumor invasion, treatment resistance, and metastasis [56]. Tumors overexpressing hypoxic markers such as HIF-1α and CAIX are associated with worse outcomes [57]. CAIX is suitable for evaluating long-term hypoxia in tumors and serves as a reliable prognostic marker [58]. In contrast, HIF-1α is better suited for assessing real-time dynamic changes in hypoxia [58], particularly in basic research aimed at elucidating the molecular mechanisms underlying hypoxia. However, HIF-1α detection is less stable, requires more stringent experimental conditions, and exhibits lower specificity compared to CAIX [58]. In our study, the lack of significant correlations between radiomic features and CAIX expression levels may primarily be attributed to the high vascularization of osteosarcoma. In addition, CAIX expression is focal and typically localized to necrotic or hypoxic zones, which are often excluded during specimen sampling to avoid non-viable tissue [59]. In contrast, radiomic analysis contains the entire tumor region. The inherent heterogeneity of

tumors further complicates the correlation between hypoxia markers and radiomic features, as various tumor regions may exhibit varying levels of hypoxia [60].

Our study also has some limitations. First, while the sample size of this study is larger than that of some previous studies, it still appears relatively small for a multicenter study. A larger and more diverse cohort could enhance the generalizability of the findings. Second, the external test sets had low preoperative NAC rates (14.3% and 58.3%), which may influence model generalizability given the prognostic importance of chemotherapy response. Thus, a subgroup analysis based on NAC or non-NAC was conducted, suggesting that the models showed consistent predictive accuracy for both NAC (AUC = 0.82) and non-NAC (AUC = 0.89) subgroups, indicating its independence from chemotherapy effects. Third, due to the retrospective data collection, the radiomic model was inevitably affected by potential selection biases and variations in MR image acquisition. To mitigate this issue, we conducted external testing to evaluate how data heterogeneity affects model performance. Moreover, we implemented grayscale normalization to minimize variations in grayscale levels across MRI images. Fourth, we utilized only CE-FS-T1WI images to construct the radiomic model, while multi-parametric MRI combining CE-FS-T1WI and other sequences may potentially enhance model discrimination in predicting survival outcomes. Nevertheless, as per the recommendation of the METRICS scoring tool, using a single MRI sequence rather than multiple should be preferred, as multi-parametric imaging may unnecessarily increase data dimensionality, feature variability, and risk of model overfitting [19]. Fifth, the external test set 2 included a mix of centers, which might introduce variability. However, we could not separate centers into their own test sets to evaluate the robustness of the model in more distinct settings, due to the very small sample sizes in each center. Sixth, tumor segmentation was performed manually by radiologists, which is prone to inter-observer variability. Automated or semi-automated segmentation techniques with deep learning might improve consistency; however, reliable segmentation remains challenging due to the inherent nature of the tumors, such as irregular shapes and ill-defined boundaries with surrounding edema [61]. In addition, our MRI data is insufficient for training a deep learning model to automatically segment osteosarcoma lesions. Seventh, while we did not compare Relief with other feature selection methods, such as the least absolute shrinkage and selection operator and Boruta, Relief has shown satisfactory performance by effectively balancing feature relevance and redundancy. In future work, we may compare various selection techniques to further refine our experimental design. Eighth, while this study explored the relationships between radiomic features and biological processes using H&E and IHC data, further validation incorporating genomic or proteomic data could enhance the biological insights derived from radiomics. Ninth, no significant correlations were observed between the radiomic features and CAIX, which might reflect limitations in the sampling or IHC quantification. Further studies could explore alternative or additional markers (e.g., HIF-1α) that could better capture hypoxia's impact on radiomic features. Finally, despite the favorable performance of the radiomic model on two external test sets, further validation in prospective settings and non-Chinese populations should be thoroughly conducted.

Conclusion

The MRI-based radiomic model shows potential in predicting DFS in osteosarcoma patients, offering supporting evidence for personalized treatment strategies. The varying degrees of correlations observed between H&E- and IHC-derived TME biomarkers and radiomic features validate the biological underpinnings of the radiomic features, affirming their interpretability at the cellular level. This interdisciplinary approach facilitates a thorough evaluation of tumor profiles spanning from microscopic to macroscopic levels, enhancing our understanding of tumor heterogeneity and guiding the development of customized treatment strategies.

Abbreviations

DFS	Disease-free survival
H&E	Hematoxylin and eosin
IHC	Immunohistochemistry
WSIs	Whole slide images
MRI	Magnetic resonance imaging
NAC	Neoadjuvant chemotherapy
TME	Tumor microenvironment
METRICS	Methodological Radiomics Score
CE-FS-T1WI	Contrast-enhanced fat-suppressed T1-weighted images
ROI	Regions of interest
GLCM	Grey-level co-occurrence matrix
GLRLM	Grey-level run length matrix
GLSZM	Grey-level size zone matrix
GLDM	Grey-level dependence matrix
NGTDM	Neighbourhood grey-tone difference matrix
RF	Random Forest
SVM	Support Vector Machine
LR	Logistic Regression
AdaBoost	Adaptive Boosting
MLP	Multilayer Perceptron
NB	Naive Bayes
FFPE	Formalin-fixed paraffin-embedded
CD3	Pan T cells
CD8	Cytotoxic T cells
CD68	Macrophages
FOXP3	Treg cells
CAIX	Carbonic Anhydrase IX
ROC	Receiver operating characteristic
DCA	Decision curve analysis

CIC	Clinical impact curve
AUC	Area under the curve
SHAP	Shapley Additive Explanations

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12916-025-04424-0.

Additional file 1: Table S1. MR image acquisition protocol. Table S2. The extracted radiomic features. Table S3. 150 patient-level nuclear features extracted from H&E WSIs. Description of 10 types of morphological nuclear features. Table S5. The selected radiomic features. Table S6. Univariate and multivariate Cox analyses of the clinical variables.

Additional file 2: Appendix file 2: Protocol of IHC-staining

Additional file 3: Figure S1. Two representative cases of hypoxia-immune-related biomarkers. Figure S2. Distribution of Dice and ICC values for inter-observer segmentation variability. Dashed lines denote the median values, while solid lines represent the first and third quartiles (25th and 75th percentiles, respectively).

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Authors' contributions
Qiuping Ren, Xiao Zhang, Xuewei Wu access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Concept and design: Ge Wen, Xiaodong Zhang, Shuixing Zhang, Bin Zhang. Acquisition, analysis, or interpretation of data: Qiuping Ren, Heng Zhao, Yongxin Zhang, Yubin Yao, Yinping Leng, Xiaoyang Zhang, Yumeng Liu, Jijie Xiao, Wenwen Liu, Xia Xie, Nana Pei, Rongfang He, Na Tang. Drafting of the manuscript: Qiuping Ren, Xuewei Wu. Critical revision of the manuscript for important intellectual content: Bin Zhang. Statistical analysis: Qiuping Ren, Xiao Zhang, Xuewei Wu. Obtained funding: Shuixing Zhang, Bin Zhang. Administrative, technical, or material support: Shuixing Zhang, Bin Zhang. Supervision: Shuixing Zhang, Bin Zhang. All authors read and approved the final manuscript.

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Data availability
The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
This study was approved by the ethics committee of the First Affiliated Hospital of Jinan University (Approval number: 20230227). Informed consent was waived because of the retrospective nature of the study and the analysis used anonymous clinical and pathological data.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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